Smart Mosquito Control System

Project ID :25-26J-369

Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor | Date

(Ms. Anjalie Gamage)

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ABSTRACT

A significant public health concern in urban Sri Lanka are mosquito-borne diseases, especially dengue and chikungunya. It is important to identify the mosquito vectors in real-time for effective control of mosquito-borne illnesses. This research project aims to help in the development of a smart mosquito control product by developing AI-Based Real-Time Mosquito Identification System. system uses a lightweight edge-deployable artificial intelligence model that processes both images and the sound of wingbeats to distinguish mosquitoes from other flying insects. identification model will leverage pre-existing open datasets for pre-training however, the model will be trained using additional data collected from local urban Sri Lankan environments. We will collect localized insect data to develop and refine our model to be relevant to the species of mosquito found in Sri Lanka. The goal of our work is to provide accurate real-time identification in order that targeted measures can be employed to control mosquito populations while reducing the impact on non-harmful species. This research supports enhanced accuracy in mosquito control and ultimately benefits the broader public health policy by generating new data to respond to preventive measures.

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1 INTRODUCTION

Mosquito-borne diseases such as dengue and chikungunya remain among the most pressing public health concerns in urban Sri Lanka. The high density of breeding sites, rapid urbanization, and changing climatic conditions have created environments where disease-transmitting mosquito populations thrive. Traditional control methods, including fogging and chemical spraying, are often inefficient and imprecise, posing risks to non-target insect species and contributing to environmental harm. The increasing resistance of mosquitoes to chemical insecticides further limits the long-term effectiveness of these conventional approaches.

To overcome these challenges, modern technologies such as artificial intelligence and edge computing offer new opportunities for precision mosquito identification and control. Advances in computer vision and bioacoustics analysis now make it possible to distinguish mosquitoes from other flying insects in real time. When coupled with IoT-enabled devices, such systems can deliver accurate, localized monitoring that supports smarter and more sustainable mosquito management.

However, most existing solutions are hindered by two critical shortcomings: lack of precision in distinguishing target mosquito species and limited adaptability to specific regional contexts. In particular, the absence of localized datasets tailored to Sri Lankan mosquito species has restricted the accuracy of AI models trained on general insect datasets.

This research addresses these gaps by proposing an AI-Based Real-Time Mosquito Identification System. The system employs a lightweight, edge-deployable artificial intelligence model that analyzes both images and wingbeat sound to distinguish mosquitoes from other insects. To improve accuracy and relevance, localized insect data will be collected and incorporated into model training, ensuring the system is specifically adapted to Sri Lanka’s mosquito species. By enabling selective and real-time mosquito identification, this approach reduces harm to non-target species while providing accurate data to guide targeted mosquito control efforts. Ultimately, the system contributes not only to precision mosquito management but also to broader public health initiatives aimed at reducing the spread of dengue and chikungunya.

1.1 Background & Literature Survey

Mosquito borne diseases, such as dengue and chikungunya, remain one of the most significant public health threats facing Sri Lanka, especially in urban areas where populations are dense. The rapid urbanization, availability of breeding sites, and climate changes have created an environment in which mosquitoes are flourishing. Traditional control measures fogging, and chemical spraying are common in practice, but their effectiveness has been declining. There are methods of vector control that are imprecise, harmful to non-target species, and problematic for the environment. Even worse, many mosquitoes are developing resistance to insecticides, meaning that in the future these approaches will also not be viable [1]. Therefore, there is an urgent need for selective, sustainable, and technology-based methods of vector control.

Recent advances in artificial intelligence (AI) and sensor technology offer exciting opportunities for automated mosquito detection and classification. The image-based methodologies for classifying mosquito species have progressed significantly. Examples of classification accuracy using convolutional neural networks (CNNs) are impressive, and some studies that used networks like AlexNet recorded classification accuracy above 90% [2]. More recent models, such as vision transformers), have achieved accuracy above 99% [3]. High accuracy can also be achieved based on lightweight architectures like MobileViT, which shows a high classification accuracy on devices with limited computational resources; this architecture is suitable for edge deployment. There are also self-supervised learning approaches with accuracy above 96%, which have the advantages of less reliance on large, annotated datasets, and still demonstrating high accuracy [4]. Based on these results, we can infer that image-based models will be able to deliver an identification with reliable and valid real-time identification, provided that we have the right dataset.

In addition to computer vision, bioacoustics has also surfaced as a complementary technique for mosquito identification. Mosquitoes produce wingbeat sounds with species-specific frequency functions that can also be gathered and analyzed using deep learning. Research found that convolutional neural networks can be utilized on spectrograms of recorded wingbeats and obtain classification accuracy rates exceeding 97% on Aedes aegypti mosquitoes [5]. Mobile phone's acoustic sensing applications have been developed to allow communities to contribute to surveillance with recorded wingbeat sounds [6]. With that said, wingbeat frequency functions do depend on age, sex of mosquito, and the conditions in the environment which means that the reliability of acoustics use in isolation is limited[7].

To overcome these challenges, hybrid techniques combining image-based features with acoustic signals are gaining traction. Research has shown multimodal models can more reliably classify mosquitoes as mosquitoes instead of other flying insects and can do so in different contexts (an environmental aspect) [8]. Open datasets such as WINGBEATS and the MACSFeD provide useful training data [9], although they are predominantly based on non-local species. In countries such as Sri Lanka, the absence of local insect identification data remains a significant constraint to using accurate models.

Moreover, experiments using IoT-enabled mosquito traps with AI models that can be integrated into real-time monitoring systems have been conducted. The traps can classify mosquitoes in a real-time setting and upload the results as data is collected to cloud-based dashboards to visualize mosquito surveillance data [10]. While these developments are encouraging, most of the prototypes remain reliant on high-power hardware and are not designed for low-resource settings (i.e., field deployment).

Although there have been advancements in AI-based mosquito identification, major gaps in research exist. Current systems are primarily not suited to local ecological settings, cannot be deployed lightweight or easily, and/or cannot utilize multiple sensing modalities. This project will fill these gaps by developing a lightweight, edge-deployable AI model that has image recognition as well as wingbeat sound analysis. The proposed system will tailor existing open datasets with local datasets collected from Sri Lankan environments and will identify local species in real-time. This select, reliable, and good information is the basis for smarter mosquito control systems and will contribute directly to public health initiatives aimed at combating dengue and chikungunya.

1.2 Research Gap

Although we have come a long way with AI-based mosquito classification, there are significant gaps that hinder the applicability of existing solutions in practice, especially in urban Sri Lanka. Most existing work assigns classes to insects based on generalized insect datasets, like WINGBEATS and MACSFeD [9], which do not specify the inherent bias towards non-local species. This in turn limits their usefulness in differentiating Sri Lankan mosquito species from other, similar non-target insects.

Further complicating the issue, many existing solutions rely predominantly on either image classification or analysis of wingbeat sound patterns, as opposed to combining either. Image recognition models CNNs [2], vision transformers [3] can achieve high quality accuracy result; however, many models require significant computing resources, rendering them unsuitable for lightweight edge-deployable devices. Similarly, acoustic-based models [5], [6] are susceptible to background noise interference, in addition to their vulnerability due to effects of wingbeat frequency associated with age, sex, and ecological circumstances [7], that affect their reliability when used alone.

Most AI-capable mosquito traps or monitoring systems [10] are designed for use in a laboratory or controlled setting rather than the field. They typically require high-powered hardware with continuous cloud-based access to the internet which would be impractical for wide-scale deployment in urban environments with limited resources.

To date, there are limited multimodal systems that combine image data and sound data for increased precision and whose deployments on edge are considered optimal. Even though hybrid systems [8] show promise, they are seldom tested in field settings and unsuited for regional species. In the Sri Lankan case, the lack of locally adapted insect datasets [1] limits the possible construction of precise, and real-time, mosquito identification systems.

This research seeks to overcome these limitations by developing an edge-deployable AI model, that combines image recognition and wingbeat sound recognition, in fields ubiquitous to urban Sri Lankan environments, in real time, and adapted for use for buzz-based recognition of actual local mosquito species- that is to mis-identify mosquitoes when outfitted correctly. The aim is to provide accurate, real-time, mosquito identifications to assist with targeted, eco-friendly vector control, creating a hurry-up way to reduce the barriers in a neglected research and applied field.

Table 1: High-Level Representation of the Research Gap

Feature / Method	**CNN-based Models [2]**	**Vision Transformers [3]**	**Acoustic CNN [5]**	**Hybrid Approaches [8]**	**IoT Mosquito Traps [10]**	**Proposed Research**
Image-based classification	✓	✓	✗	✓	✓	✓
Acoustic-based classification	✗	✗	✓	✓	✓	✓
Multimodal (Image + Sound)	✗	✗	✗	✓	✗	✓
High accuracy in controlled labs	✓	✓	✓	✓	✓	✓
Edge-deployable / Lightweight	✗	✗	✗	✗	✗	✓
Uses localized Sri Lankan data	✗	✗	✗	✗	✗	✓
Real-time field deployment	✗	✗	✗	✗	✗	✓

1.3 Research Problem

Despite progress with artificial intelligence, IoT-based mosquito surveillance, and improving techniques to collect mosquitos, there are still some key challenges to be addressed in a Sri Lankan context. Mosquito identification techniques, whether based on images or sound, have been able to discriminate mosquitoes within the laboratory, but have not been able to do so in a more natural urban environment, where there are significant levels of environmental noise, variable lighting, and non-target insect species.

Most of the models are built for general datasets of insects, but mosquitos are not representative in datasets as other species in Sri Lanka. Unfortunately, there are not enough quality datasets to develop AI models and deploy and derive results in Sri Lanka. This lack of localized data means the AI models struggle with recognizing harmful from harmless infection vectors, meaning less accuracy in directing swarming targeted mosquito management.

Another area for consideration is the computational intensity of existing solutions. Many of the models currently applying deep learning are compute-heavy and do not run on low-power edge devices, which are needed for real-time on-site application in mosquito monitoring activities. Without electricity-free rated, optimized models that are cost-effective, implementing practical mosquito monitoring in cost-constrained environments is unachievable.

Additionally, existing methods to classify insects and birds either rely solely on image recognition or wingbeat frequency. These methods are adversely impacted with variations in the environment and do not always produce consistent results in noisy and complicated environments. Also, while a multimodal approach combining both visual and acoustic features has yet to be investigated and explored by researchers, it can provide greater reliability and accuracy.

2 OBJECTIVES

2.1 Main Objectives

The main goal of this research is to design and deploy a lightweight, edge-deployable AI-based real-time mosquito identification system to accurately identify mosquitoes that spread disease from non-target insects and utilizes multimodal inputs of data (image analysis / wingbeat sound analysis) in real-time. In addition, the system will be adapted to the species of mosquitoes found in Sri Lanka to ensure accuracy and appropriateness. In doing so, this project will offer a reliable technological basis for selective mosquito control and subsequently reduce the risk of transmission of dengue and chikungunya in urban populations.

2.2 Sub-Objectives

Data Collection and Preparation

- Collect localized datasets of mosquito images and wingbeat sounds from urban Sri Lankan environments.
- Preprocess and annotate the data to ensure quality and consistency for model training.

Model Development and Optimization

- Design and train a lightweight, edge-deployable AI model capable of distinguishing mosquitoes from other flying insects.
- Optimize the model for real-time processing on resource constrained IoT devices.

Integration of Multimodal Features

- Incorporate both computer vision (image-based) and bioacoustics (wingbeat frequency) features to enhance classification accuracy.
- Evaluate the contribution of each modality individually and in combination.

System Implementation and Deployment

- Implement the AI model on an IoT-enabled edge device for real-time mosquito identification.
- Ensure low-latency processing and adaptability to different urban environments.

Performance Evaluation

- Test the system against benchmark datasets and localized field data to measure accuracy, precision, recall, and response time.
- Compare performance with existing mosquito detection approaches.

Public Health Relevance

- Generate actionable mosquito population data that can support vector control strategies.
- Ensure minimal harm to non-target insect species to promote ecological balance.

3 METHODOLOGY

3.1 Overall System

The proposed solution is a smart AI-integrated IoT framework designed to support selective mosquito control and strengthen public health monitoring in urban Sri Lanka. The methodology follows a modular approach, where each component contributes to achieving accurate identification, effective trapping, and real-time surveillance of vector-borne disease risks. The system consists of four major modules:

- **AI-Based Mosquito Identification**

  This module applies advanced AI models to distinguish disease-carrying mosquito species from harmless insects. Captured mosquitoes are analyzed through image recognition and wingbeat sound analysis. A lightweight, multimodal architecture combining computer vision and acoustic classification ensures accurate species detection even under varying environmental conditions.

- **Smart Trap Mechanism**

  This module employs an eco-friendly trapping system that mimics human attractants, such as UV light, controlled CO_2 release, and heat emission, to lure mosquitoes. Once captured, insects are stored in a containment chamber where they are scanned by the AI-based identification unit. Non-target insects are safely released, while harmful species are logged and flagged for further analysis.

- **IoT Sensor Network and Cloud Dashboard**

  This module integrates IoT-based environmental sensors, including temperature, humidity, and CO_2 detectors, to monitor factors influencing mosquito activity. Data collected from traps and sensors are transmitted to a centralized cloud platform. A real-time dashboard visualizes mosquito density maps, environmental conditions, and temporal patterns, enabling health authorities to track potential breeding hotspots.

- **Health Data Integration and Alert System**

  This module incorporates hospital-reported dengue and chikungunya cases into the cloud system. By combining environmental and entomological data with health statistics, the system generates outbreak heatmaps. A public-facing web and mobile application provides timely alerts to residents and notifies public health officials for targeted interventions in high-risk zones.

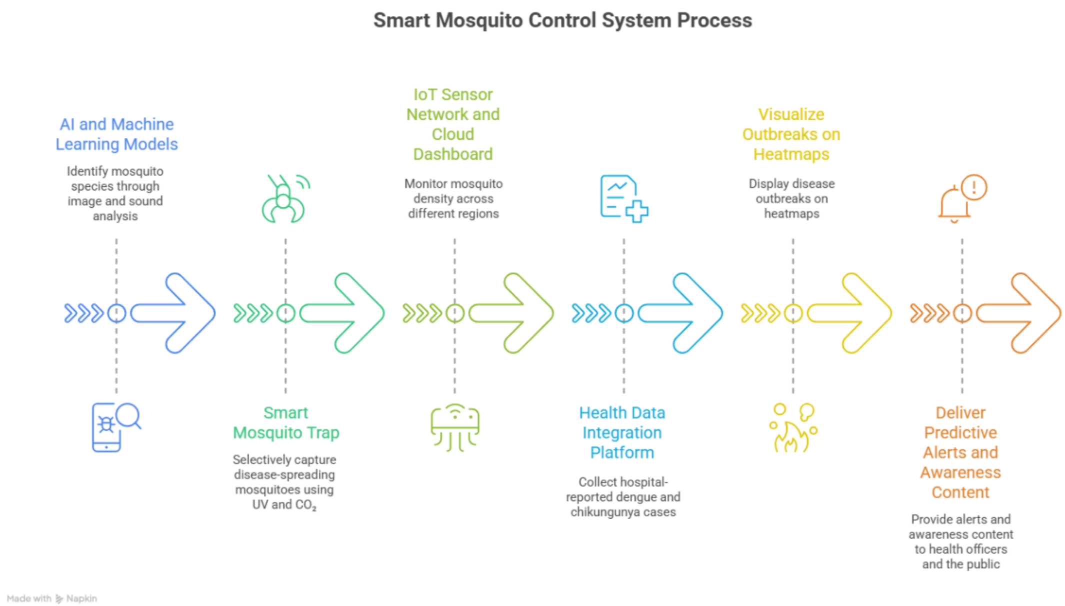


Figure 1 : Smart Mosquito Control System Process

3.2 Individual System

This component is responsible for the accurate identification of mosquito species captured within the smart trap. Its primary role is to ensure that only disease-spreading mosquitoes, such as *Aedes aegypti* and *Aedes albopictus*, are flagged, while harmless insects are excluded. The methodology for this individual system is structured as follows:

- **Data Collection**
    - Acquire a localized dataset of mosquito species common in Sri Lanka, using both high-resolution images and wingbeat sound recordings.
    - Environmental variations (daylight, low-light, background noise) will be included to simulate real-world urban conditions.
    - Data will be annotated and categorized into disease-spreading and non-harmful species for supervised training.

- **Preprocessing**
    - Images will be processed through normalization, resizing, and augmentation techniques to improve robustness under varying light conditions.
    - Wingbeat recordings will be filtered to reduce environmental noise and converted into spectrograms for feature extraction.

- **Model Development**
    - Design a lightweight multimodal AI model combining Convolutional Neural Networks (CNNs) for image recognition and Recurrent Neural Networks (RNNs) for wingbeat sound analysis.
    - Implement multimodal fusion to integrate visual and acoustic features, increasing classification accuracy in noisy environments.

- **Optimization for Edge Devices**
    - Apply model compression techniques such as quantization and pruning to reduce memory usage and processing requirements.
    - Deploy the optimized model on low-power edge devices to support real-time operation inside the mosquito trap.

- **System Integration**
    - The trained model will be embedded into the smart trap unit, directly processing data from the trap’s camera and microphone sensors.

- Identification results will be transmitted to the IoT sensor network and cloud dashboard for further analysis and reporting.

- **Evaluation**
  - The system will be tested in both controlled laboratory conditions and real urban environments.
  - Performance metrics such as accuracy, precision, recall, latency, and power consumption will be measured.
  - Results will be compared against existing single-modality identification methods to validate improvements in robustness and efficiency.

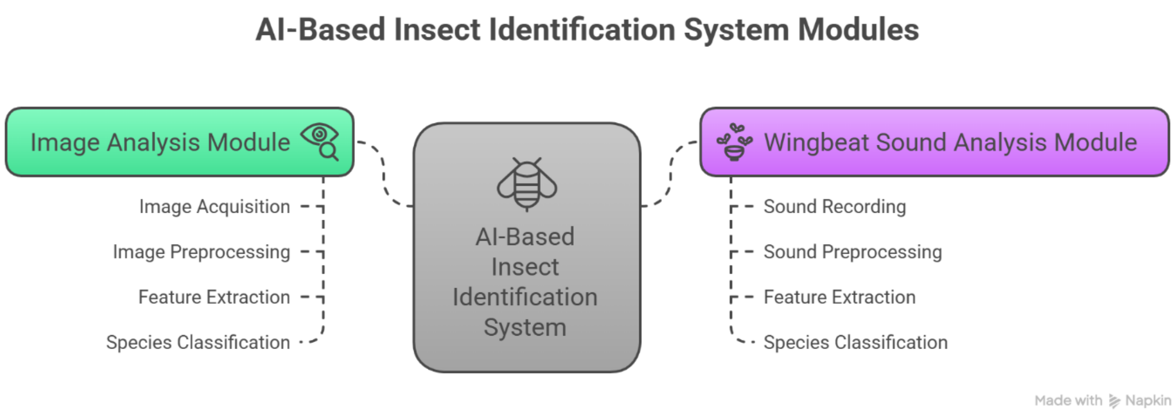


Figure 2 : AI based Mosquito Identification System

4 PROJECT REQUIRMENTS

4.1 Functional Requirements

These define what the system must do to achieve its objectives:

1. **Real-Time Mosquito Detection**
    - Capture images of flying insects using the edge camera.
    - Detect and identify mosquitoes among other insects in real time.
2. **Wingbeat Sound Analysis**
    - Record insect wingbeat sounds.
    - Analyze audio to distinguish mosquito species using frequency patterns.
3. **Data Collection and Storage**
    - Collect images and audio of mosquitoes for training and continuous model improvement.
    - Store detection results with timestamps locally on the edge device.
4. **Edge AI Model Deployment**
    - Run a lightweight AI model capable of real-time inference on an edge device.
    - Support both image-based and sound-based mosquito classification.
5. **Integration with IoT System**
    - Communicate detected mosquito counts and species to the central IoT system.
    - Optionally trigger eco-friendly traps or notifications based on detection.
6. **User Interface (Optional)**
    - Display real-time mosquito detection statistics locally or on a connected dashboard.

4.2 Functional Requirements

These define how the system **performs** and the constraints it must satisfy:

1. **Performance and Speed**

   - Real-time detection: less than 1-second processing time per insect.
   - High detection accuracy: aim for at least 90% precision in distinguishing mosquitoes from non-target insects.

2. **Scalability**
   - Able to integrate multiple edge devices in a sensor network for broader monitoring.

3. **Portability**
   - Lightweight and low-power device suitable for deployment in urban areas.

4. **Robustness and Reliability**
   - Function under varying environmental conditions: sunlight, rain, wind, and urban noise.
   - Maintain detection accuracy despite background noise in audio analysis.

5. **Security and Privacy**
   - Ensure collected data is limited to mosquitoes and insects only; no personal human data.
   - Secure communication with central systems (if connected).

6. **Maintainability and Upgradability**
   - Easy to update AI models with new mosquito data.
   - Modular system design for hardware or software component replacement.

5 GANTT CHART & WORK BREAKDOWN STRUCTURE

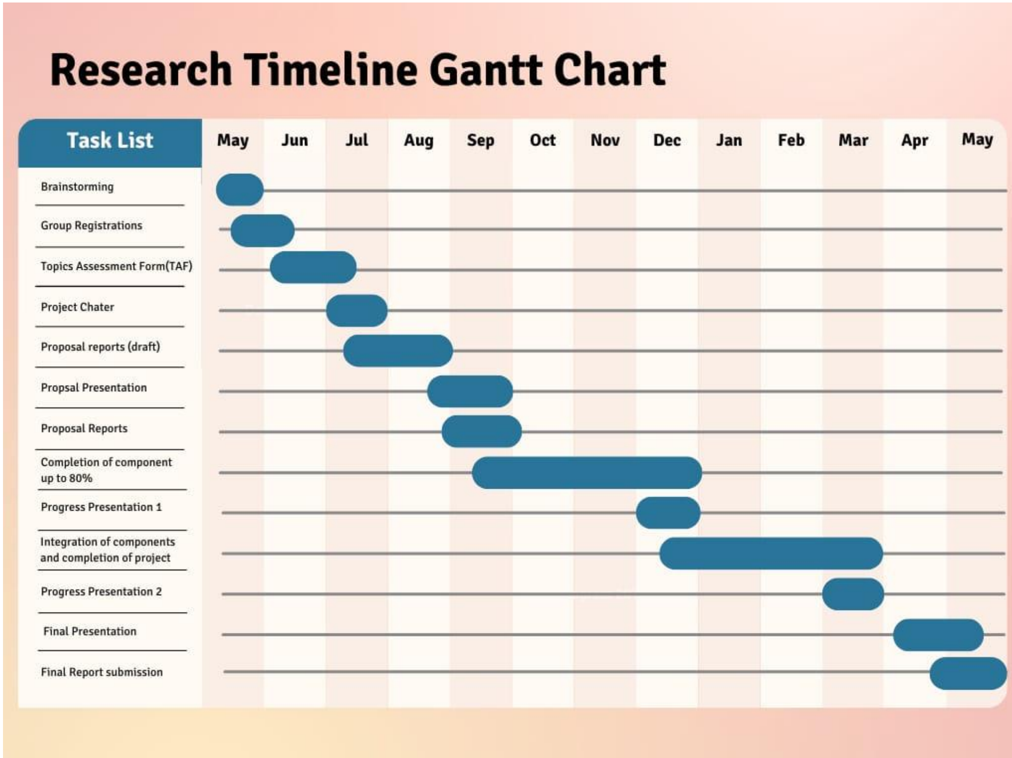
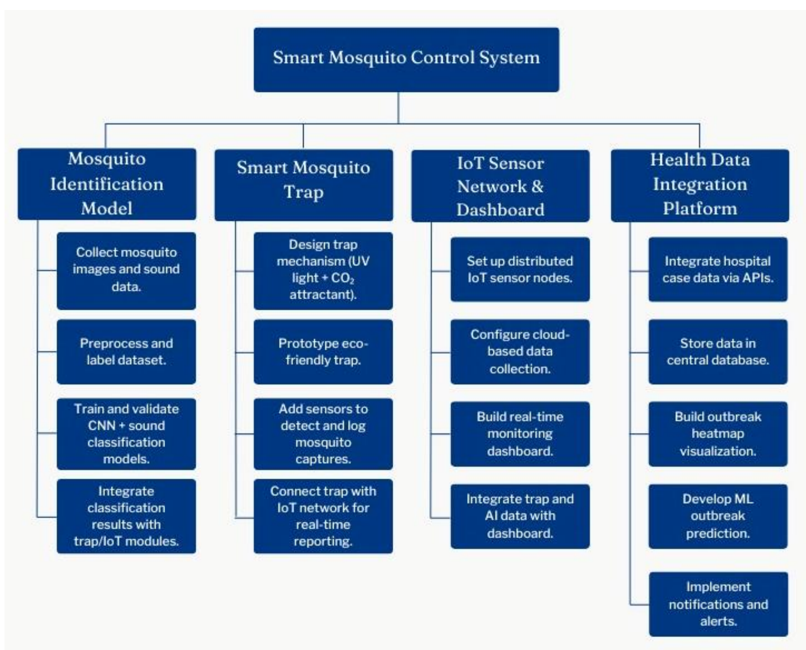


Figure 3: Gantt Chart



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7 APPENDICES